CCCA Meta-Atlas · Cell-State Analysis

Peng et al. 2019 — Pancreatic Ductal Adenocarcinoma (10x) · 7,945 malignant cells · 16 samples

GO / NO-GO GATES

Gate 1	GO	Batch mixing + continuity. All 16 samples intermix on one continuous connected component (0 detached islands; UMAP aspect 1.17; iLISI median 1.0->5.3 of 16 at theta=10) — judged on the embedding, not a scalar. Cell cycle (6.6% cycling) is not a module confound (|rho|<=0.24; the pSom-Crest spatial anti-correlation is unchanged when cyclers are excluded), so the whole malignant manifold is carried forward.
Gate 2	NO-GO	Embryonic module pattern absent. The three smoothed modules occupy a 2-D triangle (PCA variance 0.55/0.40; Gast/pSom/Crest top-decile peaks 110-134 deg apart) — no Gast/pSom co-expression hub. They do not order along pseudotime (pSom rho=+0.11, Crest rho=-0.10) and Crest is non-terminal (Crest-high decile at pseudotime 0.60; joint pSom&Crest enrichment 0.41). Neither the branching hub nor the non-branching pSom->Crest axis reproduces.
Gate 3	NA	Branching-only gate. The manifold is non-branching (curve: 2 tips, 0 forks) and Gate 2 did not pass — cross-repression is not evaluated.

HUB / r

No hub — Gast and pSom are anti-correlated and sit at opposite triangle corners (spatial r=-0.41), so no Gast/pSom-overlap convergence population exists. Hub r = NA (branching-only).

GEOMETRY

Non-branching manifold, but the embryonic modules do not resolve onto it — a triangular, non-axial module geography, not the Linear-pSom/Crest developmental axis seen in the same-tissue Steele/Lin/Moncada PDAC GO runs.

EMT-TFs

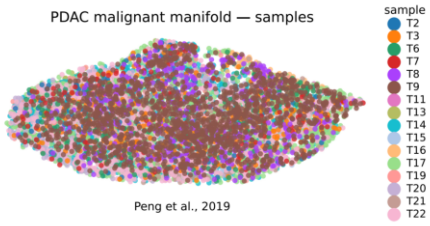
Data-derived along pseudotime (Snail/Zeb/Twist + FOXC/PRRX panel, 3% detection floor): ZEB1 (root end) and SNAI2 (terminus end). Shown unconditionally as the final notebook (04).

VERDICT

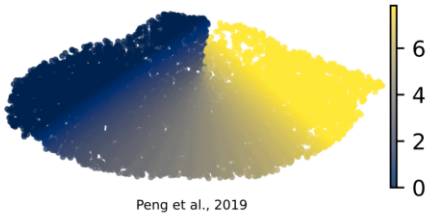
Pattern does NOT reproduce. G1 GO / G2 NO-GO / G3 NA — developmental programs are present as scores but geographically unresolved (a triangle, not an axis). Threshold: >=100-cell floor (gap-finding premise unmet), all 16 samples kept. Root: Gast/pSom-high curve tip.

STEP FIGURES

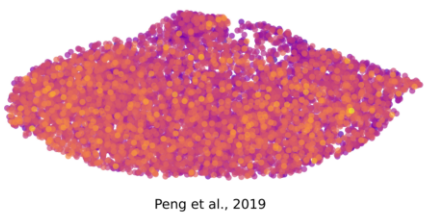
01 · samples (Gate 1)



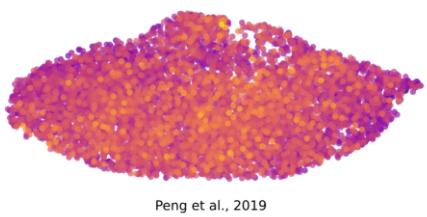
02 · pseudotime



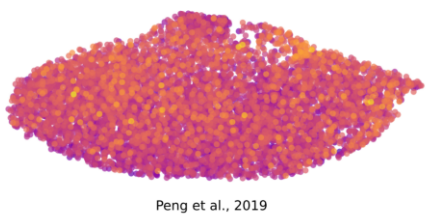
03 · Gastrulation



03 · Somitogenesis



03 · Neural crest



04 · EMT-TFs

